Virtual Pet

Assignment
Build a state machine that brings a digital character to life. Create an interactive virtual pet. There are no requirements for the type or form of the pet. We expect a project that is as unique and creative as possible.

Requirements

- Infinite Loop: The game runs indefinitely with no "Game Over" or end state.
- States & Parameters: Include at least two FSM states and two core parameters. Parameters must directly influence the pet's behavior and state transitions.
- Player Interaction: Provide a range of user interactions. Make player choices feel meaningful by ensuring different actions lead to distinctly different outcomes.
- Validation: Validate your design through playtesting. Players should experience the pet as a responsive character with its own distinct personality.

Visual design

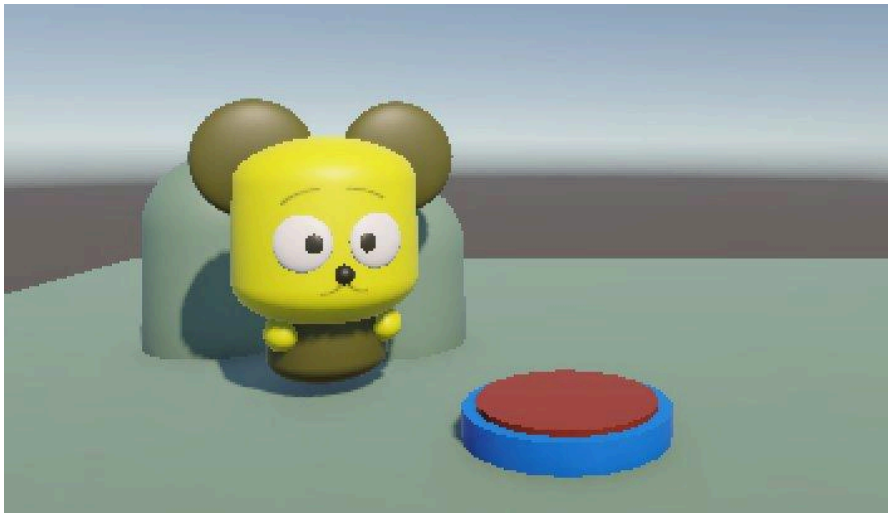
The pet must be built using primitives. You may use existing prop models, if their licenses allow it, but you may not use existing assets for the pet itself. We want you to practise creating animations in Unity, so all animations must be created by you using the Animation window.

Learning Objectives

- Work with state machines and variables.
- Create a dynamic system.
- Design and build an NPC with a clear personality.
- Apply theoretical principles to your design, such as baby schema, colour theory, and shape language.

Deliverables

- A WebGL build, fully playable without any external instructions on itch.io.
- The project files (all files, without the \Library folder) as a compressed folder.
- The design report (see below)



Design report

The document should contain the following elements:

- *A description of the pet's personality:* describe the pet's personality: what does your pet like or dislike? Is your pet serious or playful? Does it respond to certain items? What way will it respond to the way the player treats it? What does it do when idle? Provide a detailed description of its personality. Also consider how these characteristics are expressed through the pet's interactions with the player.
- *A rationale for the pet's design:* explain how the pet's visual design is derived from its personality. Discuss how elements such as shape, colour, proportions, materials, and other visual characteristics reinforce or communicate the pet's personality traits.
- *A state diagram of the pet:* create a state diagram showing the pet's different states, including its corresponding actions and emotions. Also show which events or circumstances cause the pet to transition from one state to another.
- *A summary of the playtest results and a description of the iterations:* describe the key findings from the playtest. What feedback did you receive, and what observations did you make? Then explain what changes you made to the pet based on these findings and why.

Resources

- Starter Unity Framework: https://github.com/acdh2/VirtualPet
  Note: This requires Unity version 6000.1.2f1 or above

Grading

The pet is graded using these criteria in the rubric:

	Sufficient (required)	Good	Excellent
Virtual Pet	The pet has a clearly designed personality. It is built from basic primitives. The visual design demonstrates a basic understanding of baby schema, colour theory, and shape theory. The pet has a working state machine. The user can interact with the pet in *two/three** distinct ways. The system has at least *two/three** distinct parameters, each of which triggers distinct behavioural responses. The pet has basic animations for each state. The pet has been playtested to verify that players perceive its personality and that the interactions are engaging. **=diff. for solo/duo teams*	At least one of the excellent criteria is met.	Two or more of these criteria are met: ● The pet shows emotional development throughout gameplay, with its internal state changing over time. ● The pet is deliberately designed to encourage players' caretaking behavior. ● The pet's audiovisual design is refined and cohesive, creating a strong and consistent sense of personality. ● The pet includes procedural or adaptive elements, with parameters that

			significantly influence its personality or appearance.